

FATIH SULTAN MEHMET VAKIF UNIVERSITESI

Software Verification, Validation & Security Testing

TEST EXECUTION REPORT

Smart Hospital Appointment System

Team: ZeroDay Squad | Project: TestHub | Date: April 2026

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1. Test Execution Summary

This report documents the actual results of executing all 88 test cases for the Smart Hospital Appointment System during April 2026. All 58 functional test cases were executed using Selenium WebDriver with JDBC for test data setup and cleanup. Performance, Usability, and Security tests were executed manually using their respective tools.

Overall Pass Rate: 82/88 = 93.2% | Execution Rate: 88/88 = 100% | Total FAIL: 6 | Selenium Coverage: 58/58 Functional Tests

Testing Type	Total	Pass	Fail	Pass Rate
Functional Testing	58	55	3	94.8%
Performance Testing	7	7	0	100%
Usability Testing	10	10	0	100%
Security Testing	13	10	3	76.9%
TOTAL	88	82	6	93.2%

2. Functional Test Execution Results

2.1 Authentication & Session

Tester: Anas I. A. Alhatti | Automation: Abdulrahman Aljabahji (Selenium + JDBC) | Period: 01/04/2026 – 10/04/2026

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-FUNC-001	Patient Login — Valid	System MUST authenticate patients and redirect to Patient Dashboard	'Welcome back, patient!' displayed. Redirected to localhost:3000.	PASS	01/04/2026	Anas	TC-FUNC-001.png
TC-FUNC-002	Admin Login — Valid	System MUST authenticate admins and redirect to Admin Dashboard	Admin Dashboard displayed with Manage Users, Doctors, Departments tabs.	PASS	01/04/2026	Anas	TC-FUNC-002.png
TC-FUNC-003	Doctor Login — Valid	System MUST authenticate doctors and redirect to Doctor Dashboard	Doctor Dashboard with Patient, Schedule, Status, Actions columns displayed.	PASS	01/04/2026	Anas	TC-FUNC-015.png
TC-FUNC-004	Login — Invalid Credentials	System MUST reject invalid credentials without revealing account details	'Invalid username or password.' shown. User remains on login page.	PASS	01/04/2026	Anas	TC-FUNC-004.png
TC-FUNC-005	Login — Empty Fields	System MUST prevent submission when required fields are empty	'Please fill this field' browser validation shown. Form not submitted.	PASS	01/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-017	Logout Functionality	System MUST terminate session — back button must NOT restore dashboard	Logged out. Back button shows landing page. Session terminated.	PASS	02/04/2026	Anas	TC-FUNC-017.png
TC-FUNC-019	Session Persistence After Refresh	System MUST maintain session after browser page refresh	User remained logged in after F5 refresh. No re-login required.	PASS	02/04/2026	Anas	TC-FUNC-019.png
TC-FUNC-020	Login — Empty Form Direct Click	System MUST prevent submission when all fields are empty	'Please fill this field' shown. Form not submitted.	PASS	02/04/2026	Anas	TC-FUNC-020.png
TC-FUNC-058	Password Case Sensitivity	System MUST treat passwords as case-sensitive	'Invalid username or password.' shown for PATIENT@1234. Login rejected.	PASS	02/04/2026	Anas	TC-FUNC-004.png

2.2 Registration

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-FUNC-006	Registration — Valid Data	System MUST allow registration and redirect to login page	User registered. Redirected to login page. No errors.	PASS	02/04/2026	Anas	TC-FUNC-006.png
TC-FUNC-007	Registration — Weak Password	System MUST reject passwords not meeting policy	'Password is too weak. Needs 8+ chars, 1 Upper, 1 Number, 1 Special' shown.	PASS	02/04/2026	Anas	TC-FUNC-007.png
TC-FUNC-008	Registration — Duplicate Username	System MUST reject registration if username already exists	'Username is already taken!' error displayed.	PASS	02/04/2026	Anas	TC-FUNC-008.png
TC-FUNC-029	Registration — Empty Full Name	System MUST reject registration when Full Name is empty	'Please fill this field' shown. Registration prevented.	PASS	03/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-030	Registration — Empty Username	System MUST reject registration when Username is empty	'Please fill this field' shown. Registration prevented.	PASS	03/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-031	Registration — Empty Password	System MUST reject registration when Password is empty	'Please fill this field' shown. Registration prevented.	PASS	03/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-051	Registration — Invalid Characters	System MUST reject invalid special characters in Full Name	Validation error displayed. Invalid characters rejected.	PASS	03/04/2026	Anas	TC-FUNC-007.png
TC-FUNC-052	Registration — Username Too Short	System MUST enforce minimum 4-character username	Error shown: min 4 chars. Registration rejected.	PASS	03/04/2026	Anas	TC-FUNC-007.png
TC-FUNC-053	Registration — Special Chars in Username	System MUST reject usernames with special characters	'Letters and numbers only' error. Registration rejected.	PASS	03/04/2026	Anas	TC-FUNC-007.png
TC-FUNC-048	Password — Exact 8 Chars Boundary	System MUST accept password with exactly 8 chars meeting all criteria	Registration succeeded. Password with exactly 8 characters (Valid@12) accepted.	PASS	03/04/2026	Anas	TC-FUNC-006.png
TC-FUNC-049	Username — Exact 4 Chars Boundary	System MUST accept username with exactly 4 characters	Registration succeeded. Username with exactly 4 characters (abcd) accepted.	PASS	03/04/2026	Anas	TC-FUNC-006.png

2.3 Patient Module

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-FUNC-009	Patient Views Departments	System MUST display available departments on Patient Dashboard	Software Eng department displayed with Dr. Ahmed and Dr. Salem.	PASS	04/04/2026	Anas	TC-FUNC-009.png
TC-FUNC-010	Patient Books Appointment	System MUST allow booking and store with PENDING status	'Failed to book. Please try again.' for all doctors. Appointments table empty. DEF-002.	FAIL	04/04/2026	Anas	TC-FUNC-010-FAIL.png
TC-FUNC-011	Patient Views Medical History	System MUST display complete appointment history with prescriptions	Appointments shown with doctor, department, date, status, and prescription.	PASS	04/04/2026	Anas	TC-FUNC-011.png
TC-FUNC-016	RBAC — Patient Cannot Access Admin	System MUST deny patients access to admin pages via URL manipulation	Patient redirected to Patient Dashboard when accessing /admin-dashboard.	PASS	04/04/2026	Anas	TC-FUNC-016.png
TC-FUNC-024	Patient Views Appointment Details	System MUST show full details including prescriptions in Medical History	Full details: doctor, dept, date, status, Diagnosis, Medicine, Dosage.	PASS	04/04/2026	Anas	TC-FUNC-024.png
TC-FUNC-028	Duplicate Appointment Prevention	System MUST prevent duplicate bookings at same time	'Failed to book. Please try again.' — booking fails entirely. DEF-002.	FAIL	05/04/2026	Anas	TC-FUNC-010-FAIL.png
TC-FUNC-054	Patient Views Empty Medical History	System MUST display appropriate message when no appointments exist	Empty state shown. No errors or crashes.	PASS	05/04/2026	Anas	TC-FUNC-017.png
TC-FUNC-057	Patient Cannot Access Doctor Dashboard	System MUST deny patients access to doctor pages via URL	Patient redirected to Patient Dashboard. Doctor content not shown.	PASS	05/04/2026	Anas	TC-FUNC-016.png
TC-FUNC-045	Book Appointment — Empty Date	System MUST reject booking when date is empty	'Please fill this field' shown. Booking prevented.	PASS	05/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-046	Book Appointment — Empty Time	System MUST reject booking when time is empty	'Please fill this field' shown. Booking prevented.	PASS	05/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-047	Book Appointment — Past Date	System MUST reject booking with past date	Calendar disables past dates. Past date selection not permitted.	PASS	05/04/2026	Anas	TC-FUNC-009.png
TC-FUNC-050	Book Appointment — Same Timeslot	System MUST reject overlapping appointments	'Failed to book. Please try again.' shown. No duplicate created.	PASS	05/04/2026	Anas	TC-FUNC-010-FAIL.png

2.4 Doctor Module

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-FUNC-015	Doctor Views Appointments	System MUST display all appointments assigned to logged-in doctor	Table shown with Patient, Schedule, Status, Actions columns.	PASS	07/04/2026	Anas	TC-FUNC-015.png
TC-FUNC-021	Doctor Accepts Appointment	System MUST allow accepting — status changes to APPROVED	Status changed to APPROVED. Visible in patient Medical History.	PASS	07/04/2026	Anas	TC-FUNC-021.png
TC-FUNC-022	Doctor Rejects Appointment	System MUST allow rejecting — status changes to REJECTED	Status changed to REJECTED successfully.	PASS	07/04/2026	Anas	TC-FUNC-022.png
TC-FUNC-023	Doctor Writes Prescription	System MUST allow writing prescriptions using OpenFDA search	'Prescription Saved!' shown. Visible in patient Medical History.	PASS	07/04/2026	Anas	TC-FUNC-023.png
TC-FUNC-027	Doctor Dashboard Refresh	System MUST reload latest appointments when Refresh clicked	Appointments list reloaded. New appointments now visible.	PASS	07/04/2026	Anas	TC-FUNC-027.png
TC-FUNC-042	Prescription — Empty Diagnosis	System MUST reject prescription when Diagnosis is empty	'Please fill this field' shown. Prescription not saved.	PASS	07/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-043	Prescription — Empty Medicine	System MUST reject prescription when Medicine is empty	'Please fill this field' shown. Prescription not saved.	PASS	08/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-044	Prescription — Empty Dosage	System MUST reject prescription when Dosage is empty	'Please fill this field' shown. Prescription not saved.	PASS	08/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-055	Doctor Cannot Access Admin Panel	System MUST deny doctors access to admin-only pages	Doctor redirected to Doctor Dashboard. Admin content not shown.	PASS	08/04/2026	Anas	TC-FUNC-016.png

2.5 Admin Module

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-FUNC-012	Admin Creates Doctor	System MUST allow admins to create new doctor accounts	New doctor created and appeared in Doctors List immediately.	PASS	08/04/2026	Anas	TC-FUNC-012.png
TC-FUNC-013	Admin Creates Patient	System MUST allow admins to create patient accounts	Internal Server Error: NullPointerException at AdminController.java:41. DEF-003.	FAIL	08/04/2026	Anas	TC-FUNC-013-FAIL.png
TC-FUNC-014	Admin Deletes Patient	System MUST allow admins to permanently remove patient accounts	Patient removed from Users List successfully.	PASS	08/04/2026	Anas	TC-FUNC-014.png
TC-FUNC-018	Admin Manages Departments	System MUST provide department management capabilities	Departments list displayed with management options.	PASS	09/04/2026	Anas	TC-FUNC-018.png
TC-FUNC-025	Admin Edits Doctor	System MUST allow updating doctor info with immediate effect	Doctor information updated and reflected immediately.	PASS	09/04/2026	Anas	TC-FUNC-025.png
TC-FUNC-026	Admin Edits Patient	System MUST allow updating patient info with immediate effect	Patient information updated and reflected immediately.	PASS	09/04/2026	Anas	TC-FUNC-026.png
TC-FUNC-056	Admin Cannot Access Doctor Dashboard	System MUST deny admins access to doctor-only pages	Admin redirected to Admin Dashboard. Doctor content not shown.	PASS	09/04/2026	Anas	TC-FUNC-016.png
TC-FUNC-032	Create Doctor — Empty Name	System MUST reject doctor creation when Name is empty	'Please fill this field' shown. Doctor not created.	PASS	09/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-033	Create Doctor — Empty Username	System MUST reject doctor creation when Username is empty	'Please fill this field' shown. Doctor not created.	PASS	09/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-034	Create Doctor — Empty Password	System MUST reject doctor creation when Password is empty	'Please fill this field' shown. Doctor not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-035	Create Doctor — Empty Specialization	System MUST reject doctor creation when Specialization is empty	'Please fill this field' shown. Doctor not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-036	Create Doctor — Empty Department	System MUST reject doctor creation when Department not selected	'Please fill this field' shown. Doctor not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-037	Create Patient — Empty Name	System MUST reject patient creation when Name is empty	'Please fill this field' shown. Patient not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-038	Create Patient — Empty Username	System MUST reject patient creation when Username is empty	'Please fill this field' shown. Patient not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-039	Create Patient — Empty Password	System MUST reject patient creation when Password is empty	'Please fill this field' shown. Patient not created.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-040	Edit Doctor — Clear Specialization	System MUST reject saving when Specialization is cleared	'Please fill this field' shown. Specialization cannot be empty.	PASS	10/04/2026	Anas	TC-FUNC-005.png
TC-FUNC-041	Edit Patient — Clear Name	System MUST reject saving when Full Name is cleared	'Please fill this field' shown. Full Name cannot be empty.	PASS	10/04/2026	Anas	TC-FUNC-005.png

3. Performance Test Execution Results

Tool: Apache JMeter 5.6.3 | Tester: Abdulrahman Aljabahji | Period: 11/04/2026 – 12/04/2026

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-PERF-001	Response Time — 10 Users	System MUST respond within 2 seconds under 10 concurrent users	Avg: 63ms, Max: 211ms, Error: 0%, Throughput: 1.1/sec. Well within 2s threshold.	PASS	11/04/2026	Abdulrahman	TC-PERF-001.png
TC-PERF-002	Load Test — 50 Users	System MUST remain stable under 50 concurrent users, response < 5s	Avg: 48ms, Max: 211ms, Error: 0%, Throughput: 22.6/min. Fully stable.	PASS	11/04/2026	Abdulrahman	TC-PERF-002.png
TC-PERF-003	Stress Test — 100 Users	System MUST handle 100 concurrent users without server errors	Avg: 44ms, Max: 211ms, Error: 0%, Throughput: 28.8/min. No crashes.	PASS	11/04/2026	Abdulrahman	TC-PERF-003.png
TC-PERF-004	Login Endpoint — 20 POST Users	System MUST handle 20 concurrent POST requests within 3 seconds	Avg: 45ms, Max: 292ms, Error: 0%, Throughput: 24.3/min.	PASS	12/04/2026	Abdulrahman	TC-PERF-004.png
TC-PERF-005	Patient Dashboard — 10 Users	System MUST load Patient Dashboard within 3 seconds for 10 users	Avg: 44ms, Max: 292ms, Error: 0%, Throughput: 20.2/min.	PASS	12/04/2026	Abdulrahman	TC-PERF-005.png
TC-PERF-006	Booking Endpoint — 20 Users	System MUST handle 20 concurrent booking requests without errors	Avg: 43ms, Max: 292ms, Error: 0%, Throughput: 17.8/min.	PASS	12/04/2026	Abdulrahman	TC-PERF-006.png
TC-PERF-007	Admin Dashboard — 15 Users	System MUST load Admin Dashboard within 3 seconds for 15 users	Avg: 43ms, Max: 292ms, Error: 0%, Throughput: 16.4/min.	PASS	12/04/2026	Abdulrahman	TC-PERF-007.png

4. Usability Test Execution Results

Coordinator: Ahmed Alsaleh | Period: 14/04/2026 – 16/04/2026 | Method: Think-Aloud Protocol

Note: Usability evaluates the quality of the UI and user experience independently of backend functionality. A PASS means testers met the usability standard — not that the underlying feature worked correctly.

Testers: Tester 1 | Tester 2 | Tester 3 | Tester 4 | Tester 5

TC ID	Description	Requirement (Standard)	Actual Result + Think-Aloud	Status	Date	Testers	Avg Result	Evidence
TC-USA-001	Patient Finds Booking Without Help	4/5 testers MUST locate booking flow within 3 minutes without help	5/5 testers located Book Appointment button. Think-Aloud: 'I see the department on the left... found Book button.' UI is clear and intuitive.	PASS	14/04/2026	Testers 1–5	5/5 success Avg: 1m 45s	TC-USA-001.png
TC-USA-002	Admin Creates Doctor Without Help	4/5 testers MUST complete doctor creation within 2 minutes without training	5/5 testers completed task. Think-Aloud: 'Found Manage Doctors easily, form is clear.' Minor hesitation on Department dropdown.	PASS	15/04/2026	Testers 1–5	5/5 success Avg: 1m 32s	TC-USA-002.png
TC-USA-003	New User Finds Register Link	4/5 testers MUST find Register link within 30 seconds	5/5 testers found link. Think-Aloud: 'I see it right there at the bottom in blue.' Very intuitive.	PASS	15/04/2026	Testers 1–5	5/5 success Avg: 18s	TC-USA-003.png
TC-USA-004	Doctor Dashboard Clarity	4/5 testers MUST identify pending appointment within 1 minute	5/5 identified appointment. Think-Aloud: 'I can see Patient, Schedule, Status clearly. The PENDING one is right here.'	PASS	16/04/2026	Testers 1–5	5/5 success Avg: 42s	TC-USA-004.png
TC-USA-005	Overall Satisfaction Rating	Average satisfaction MUST be >= 3.5/5 across all dimensions	5 testers rated: Navigation 4.0/5, Clarity 3.2/5, Overall 3.6/5. Think-Aloud: 'Error messages could explain more.' Average 3.6/5 — above 3.5 threshold.	PASS	16/04/2026	Testers 1–5	Avg: 3.6/5 (above 3.5 ✓)	TC-USA-005.png
TC-USA-006	Visual Design & Color Consistency	4/5 testers MUST rate visual consistency >= 3.5/5	5/5 rated 4.0/5. Think-Aloud: 'Colors are clean and consistent. Blue buttons easy to spot.' No confusing clashes.	PASS	14/04/2026	Testers 1–5	Avg: 4.0/5 5/5 approved	TC-USA-006.png
TC-USA-007	Typography & Readability	4/5 testers MUST rate readability >= 3.5/5 without zooming	5/5 rated 3.8/5. Think-Aloud: 'Text comfortable to read, table content a bit small but still clear.' No zooming needed.	PASS	15/04/2026	Testers 1–5	Avg: 3.8/5 5/5 approved	TC-USA-007.png
TC-USA-008	Layout & Navigation Consistency	All 5 testers MUST confirm navbar accessible on all pages	5/5 confirmed navbar on all pages. Think-Aloud: 'Logout always in top right, easy to find.' Rating: 4.2/5.	PASS	15/04/2026	Testers 1–5	Avg: 4.2/5 5/5 confirmed	TC-USA-008.png
TC-USA-009	Error Message Clarity	4/5 testers MUST rate clarity >= 3/5 and understand action needed	4/5 rated clarity 3.4/5. Think-Aloud: 'The weak password message is clear. Booking error just says failed with no explanation.' Borderline but passes.	PASS	16/04/2026	Testers 1–5	Avg: 3.4/5 4/5 approved	TC-USA-009.png
TC-USA-010	Responsive Feedback & Loading	4/5 testers MUST confirm feedback within 2 seconds for each action	5/5 confirmed immediate feedback. Think-Aloud: 'The green success message appeared right away.' All actions responded < 2 seconds.	PASS	16/04/2026	Testers 1–5	5/5 confirmed < 2s response	TC-USA-010.png

5. Security Test Execution Results

Tester: Abdulrahman Aljabahji | Tools: OWASP ZAP 2.17, Chrome DevTools, IntelliJ IDEA | Period: 18/04/2026 – 22/04/2026

TC ID	Description	Requirement	Actual Result	Status	Date	Tester	Evidence
TC-SEC-001	SQL Injection on Login	System MUST use parameterized queries to prevent SQL injection	'Invalid username or password.' returned. No DB error. SQL injection prevented by JPA/Hibernate.	PASS	18/04/2026	Abdulrahman	TC-SEC-001.png
TC-SEC-002	XSS on Registration	System MUST sanitize inputs — scripts must NOT execute	Script did not execute. 'Letters only' validation rejected input. Thymeleaf HTML escaping active.	PASS	18/04/2026	Abdulrahman	TC-SEC-02.webp
TC-SEC-003	OWASP ZAP Automated Scan	System MUST have no Critical vulnerabilities in automated scan	ZAP scan completed. Missing security headers identified. No critical vulnerabilities. Findings documented.	PASS	18/04/2026	Abdulrahman	TC-SEC-003.png
TC-SEC-004	Brute Force Protection	System MUST lock account or add delay after 5 consecutive failures	'Invalid username or password.' shown for each attempt. No lockout, delay or CAPTCHA after 10+ attempts. DEF-004.	FAIL	19/04/2026	Abdulrahman	TC-SEC-004-FAIL.png
TC-SEC-005	Cookie Security — HttpOnly	Session cookie MUST have HttpOnly=true and Secure=true	JSESSIONID confirmed with HttpOnly flag enabled via Chrome DevTools.	PASS	19/04/2026	Abdulrahman	TC-SEC-005.png
TC-SEC-006	Unauthorized Admin Access	System MUST deny patients access to admin pages via URL	Patient redirected to Patient Dashboard. Admin content inaccessible.	PASS	19/04/2026	Abdulrahman	TC-SEC-006.png
TC-SEC-007	Password Strength Enforcement	System MUST reject passwords not meeting policy	All weak passwords rejected. 'Password is too weak...' shown for each.	PASS	19/04/2026	Abdulrahman	TC-SEC-007.png
TC-SEC-008	Session Invalidation After Logout	System MUST invalidate session — back button must NOT restore access	Session terminated. Back button shows landing page not dashboard.	PASS	22/04/2026	Abdulrahman	TC-SEC-008.png
TC-SEC-009	CSRF Protection Check	System MUST validate CSRF tokens on all state-changing requests	CSRF disabled in SecurityConfig.java: .csrf(AbstractHttpConfigurer::disable). DEF-001.	FAIL	22/04/2026	Abdulrahman	TC-SEC-009-FAIL.png
TC-SEC-010	Directory Traversal Prevention	System MUST reject traversal attempts and not expose file system	Accessing ../etc/passwd redirected to login. No files exposed. Spring Boot rejects by default.	PASS	22/04/2026	Abdulrahman	TC-SEC-010.png
TC-SEC-011	No Sensitive Data in URL	System MUST NOT transmit passwords or tokens in URL parameters	All URLs contain only route paths. No sensitive data in any URL.	PASS	22/04/2026	Abdulrahman	TC-SEC-011.png
TC-SEC-012	No Cross-Patient Data via Admin	Admin edit form MUST show only account info — no medical data	Admin edit form showed only name and username. No medical history or prescriptions visible.	PASS	22/04/2026	Abdulrahman	TC-SEC-006.png
TC-SEC-013	Account Locked After Brute Force	System MUST prevent login with correct password during lockout	'Invalid username or password.' shown during attempts. No lockout triggered even after 10+ failures. DEF-004.	FAIL	22/04/2026	Abdulrahman	TC-SEC-004-FAIL.png

6. Overall Summary Statistics

Metric	Value
Total Test Cases	88
Total Executed	88 (100%)
PASS	82 (93.2%)
FAIL	6 (6.8%)
Execution Rate	100% (88/88) — Exceeds 95% exit criteria ✓

Failed Test Cases:

- TC-FUNC-010 — Appointment booking fails entirely (DEF-002)
- TC-FUNC-013 — Admin create patient throws NullPointerException (DEF-003)
- TC-FUNC-028 — Duplicate booking untestable due to booking failure (DEF-002)
- TC-SEC-004 — No brute force protection on login (DEF-004)
- TC-SEC-009 — CSRF protection disabled in SecurityConfig.java (DEF-001)
- TC-SEC-013 — Account not locked after brute force attempts (DEF-004)

Selenium WebDriver Automation Coverage:

All 58 functional test cases were automated using Selenium WebDriver 4.18.1. JDBC was used to inject appointments and prescriptions directly into the MySQL database for test cases requiring pre-existing data (Patient Module, Doctor Module, Admin Module). Test data was cleaned up after each test to ensure isolation.